

Quality evaluation after long-term storage

Samples fit for purpose ?

Quality of human body material samples for potential Use in scientific research Evaluated after been Stored for long Time at Sciensano - Fit For Purpose project

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The effects of long-term storage of human body material (HBM) samples on their future research potential are not well-known according to literature and this is particularly true for HBM samples being studied in the context of human infectious diseases, an important domain of expertise at Sciensano

Why doing this research ?

- High-quality HBM samples are needed for scientific research, but what happens to sample quality after long-term storage (?)
- The Belgian health institute Sciensano houses many collections of stored HBM samples, conform to the applicable biobank legislation
- The envisaged collections, of which several already stored for many years, can be used for future research

Objective of the QUEST-FFP project

- Evaluate the fitness of long-term stored HBM samples for their potential use in future research concerning human infectious diseases of interest to Sciensano

Methodology

- Eligible samples for this project are biobanked samples (can be of ≠ types) stored for > 1 year or pooled and previously positively analyzed for a pathogen of interest in the scope of a study or Sciensano activity
- Samples will be re-analyzed according to the assay that was applied for obtaining the initial study, diagnosis or surveillance result (using ≠ assays) and the re-testing result will be compared with the previous result
- The applied quality control biomarker evaluating fitness will be the loss of assay positivity per storage year

Foreseen outcomes of the project

- The duration of long-term sample conservation without loss of quality can be assessed, leading to the setup of guidelines on storage duration of our HBM samples, depending on the sample type and pathogen analyzed

Summary table of the literature study showing the gaps of knowledge concerning the availability of published articles (+ conclusions) on the investigation of the stability of detected pathogen load (for viruses of interest to Sciensano) in HBM samples after long-term storage (> 1 y.)

Virus	N° of relevant articles found ¹ and references	HBM sample type(s) or extracts studied ²	Conclusions on the stability of the viral load ³
Hepatitis B	2 [a, b]	Serum Plasma	Stable up to 10 y. (?) at -70°C and < 1 y. at -20°C
Hepatitis C	3 [a, b, c]	Serum Plasma	Stable up to 1 y. or more (3-5 y. ?) at -70°C / -20°C
Hepatitis D	0		?
Hepatitis E	0		?
Influenza A/H1N1	1 [d]	Nasopharyngeal swabs RNA extracts	Stable up to 3 y. or more (?) at -80°C
Measles	0		?
Mumps	0		?
Norovirus	1 [e]	Stool samples	Stable up to 7 y. at 4°C
Rubella	0		?
SARS-CoV-2	2 [f, g]	Dried blood spots RNA extracts	Stable up to 1 y. or more (?) at -80°C for extracts and 2 y. or more (?) at -20°C for dried blood spots

¹ Literature study performed in PubMed with searches using (combinations of) keywords like “virus name”, “long-term”, “storage”, “stability”, etc. Article references can be found below.

² The HBM samples (or RNA extracts) were studied on the stability of viral load using RT-PCR based assays, except for the dried blood spots (which were studied using an IgG ELISA assay).

³ Gaps of knowledge are indicated with a ? highlighted in yellow. These gaps of knowledge can be either a lack of studies, contradictory results between studies or only a limited number of studied storage timepoints (going not so far back in time) in these studies.

References

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